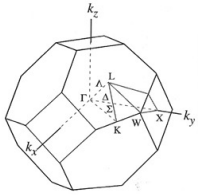
Bipolar Junction Transistors



Block Diagrams and Circuit Symbols

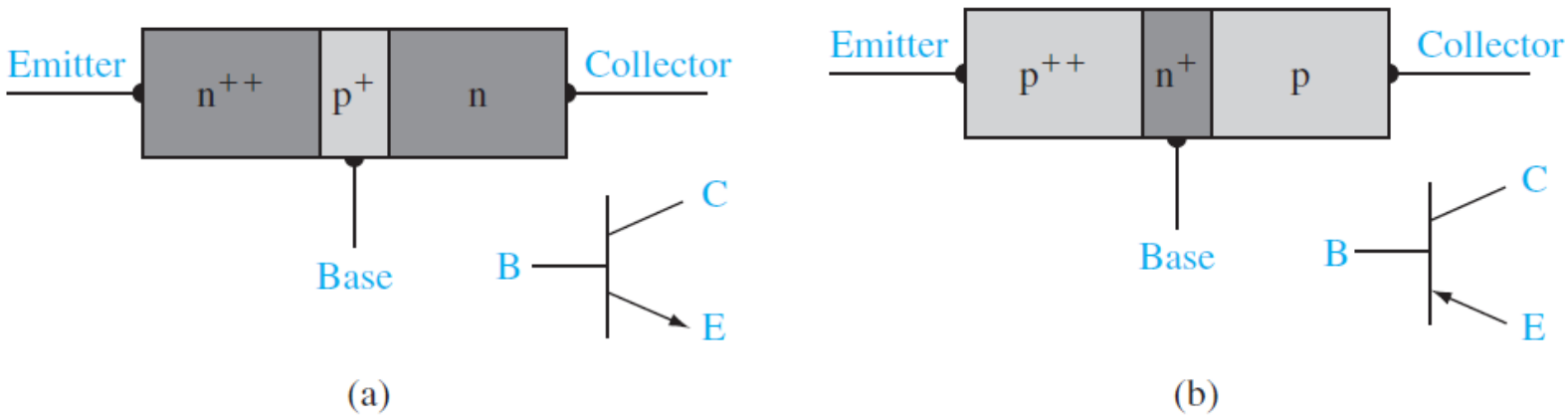
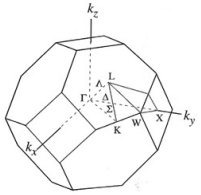


Figure 12.1 | Simplified block diagrams and circuit symbols of (a) npn and (b) pnp bipolar transistors.



Cross Section

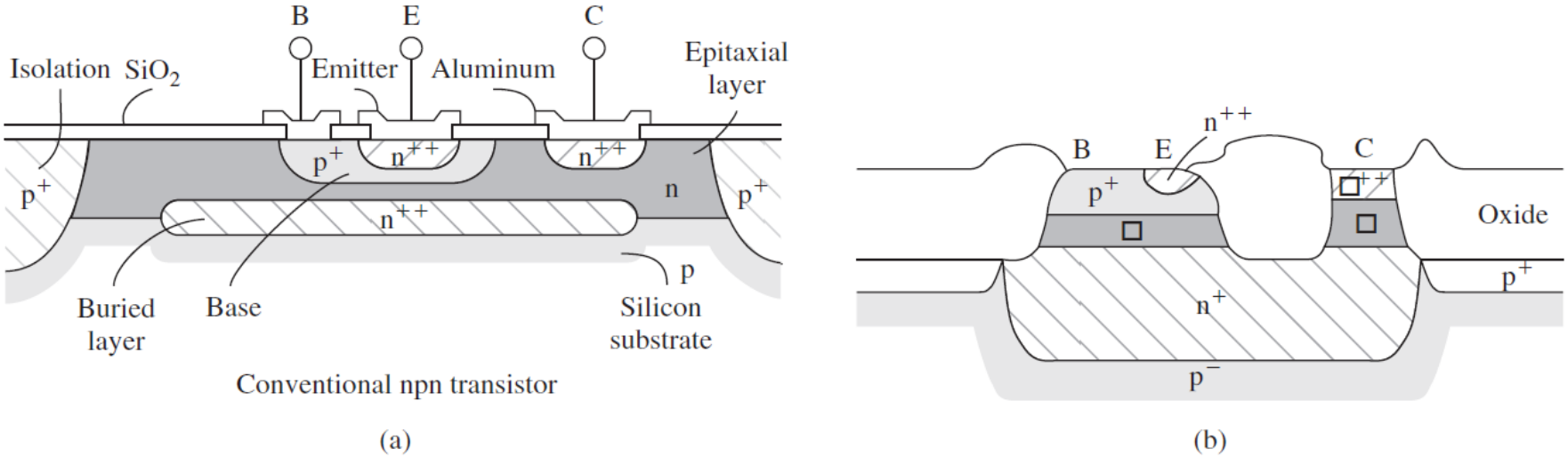
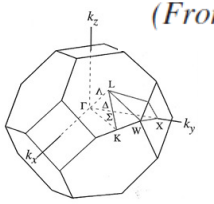


Figure 12.2 | Cross section of (a) a conventional integrated circuit npn bipolar transistor and (b) an oxide-isolated npn bipolar transistor.
(From Muller and Kamins [4].)



Basic Principle of Operation

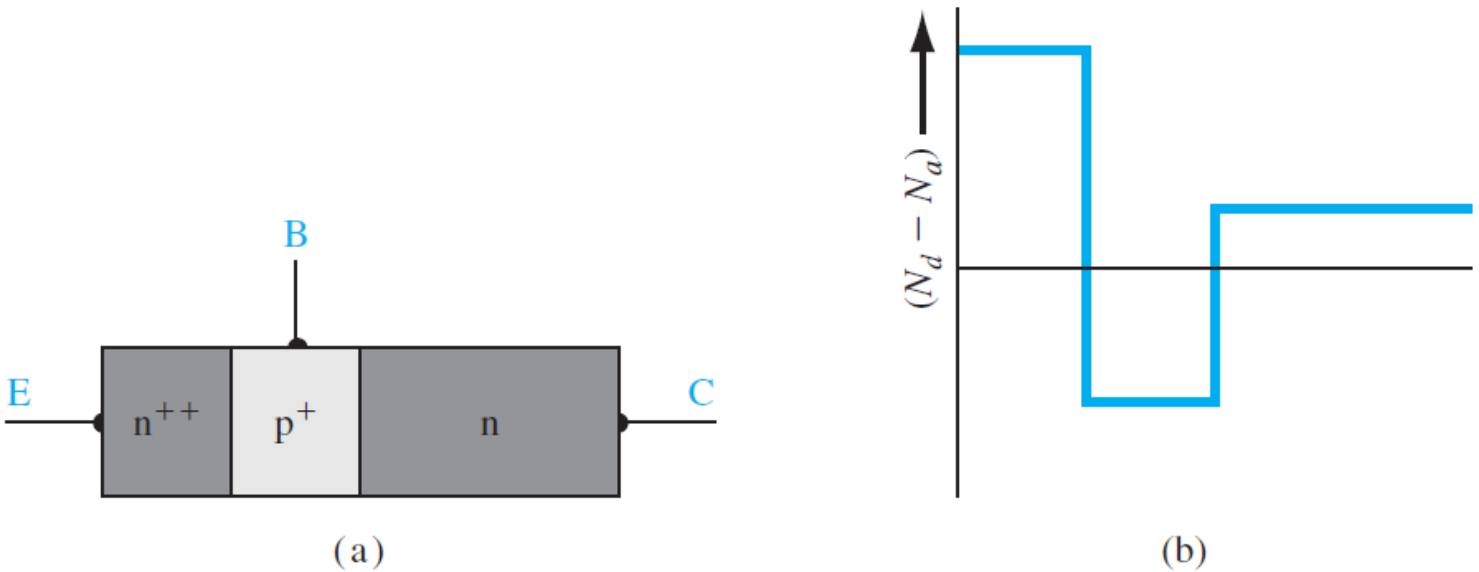
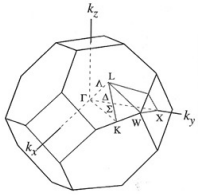
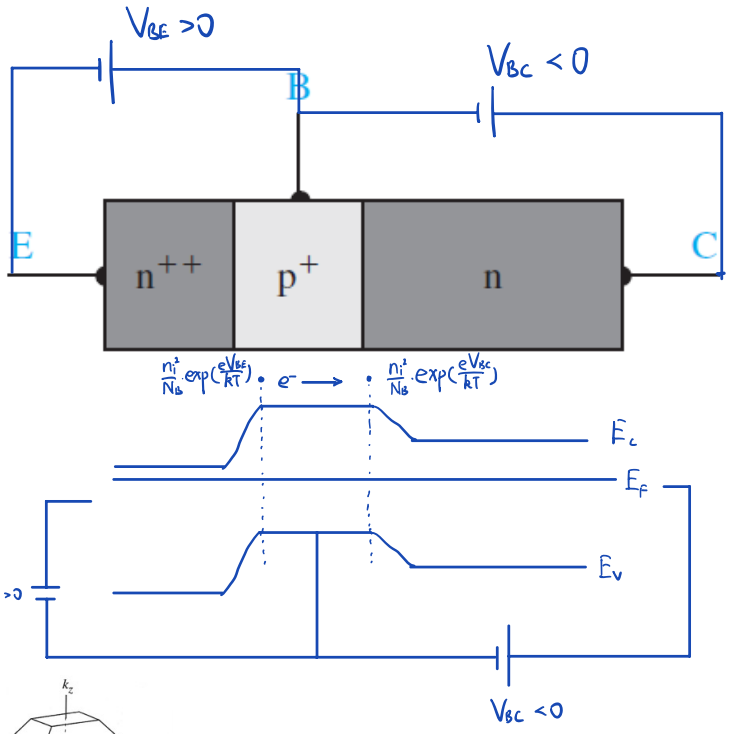
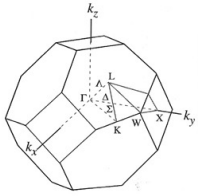


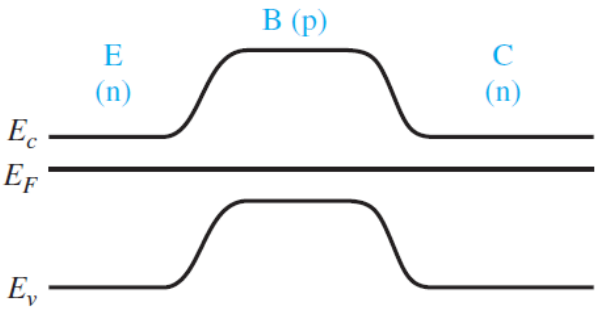
Figure 12.3 | Idealized doping profile of a uniformly doped npn bipolar transistor.



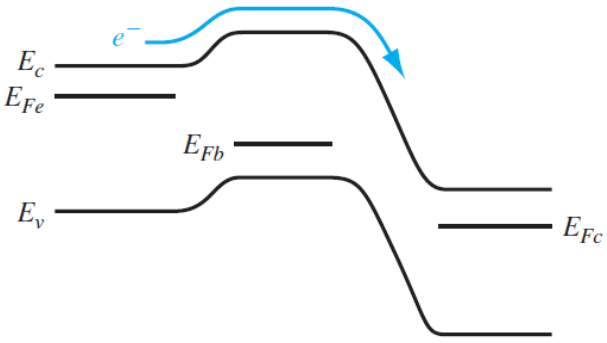
Basic Principle of Operation







Zero bias



Forward active

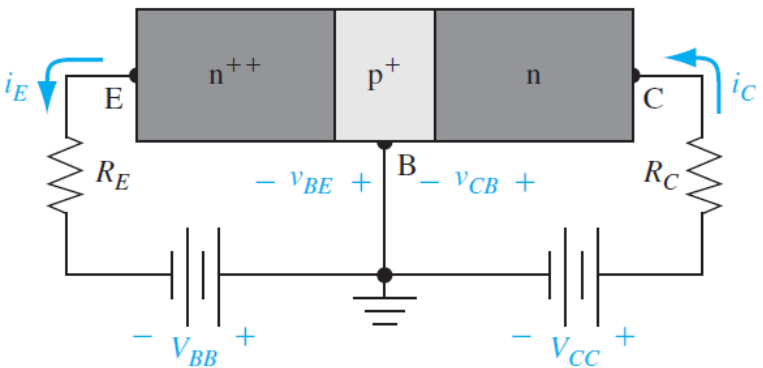
region p^+: $D_n\frac{\partial^2\delta n_p}{\partial x^2}-\frac{\delta n_p}{\tau_n}=0$

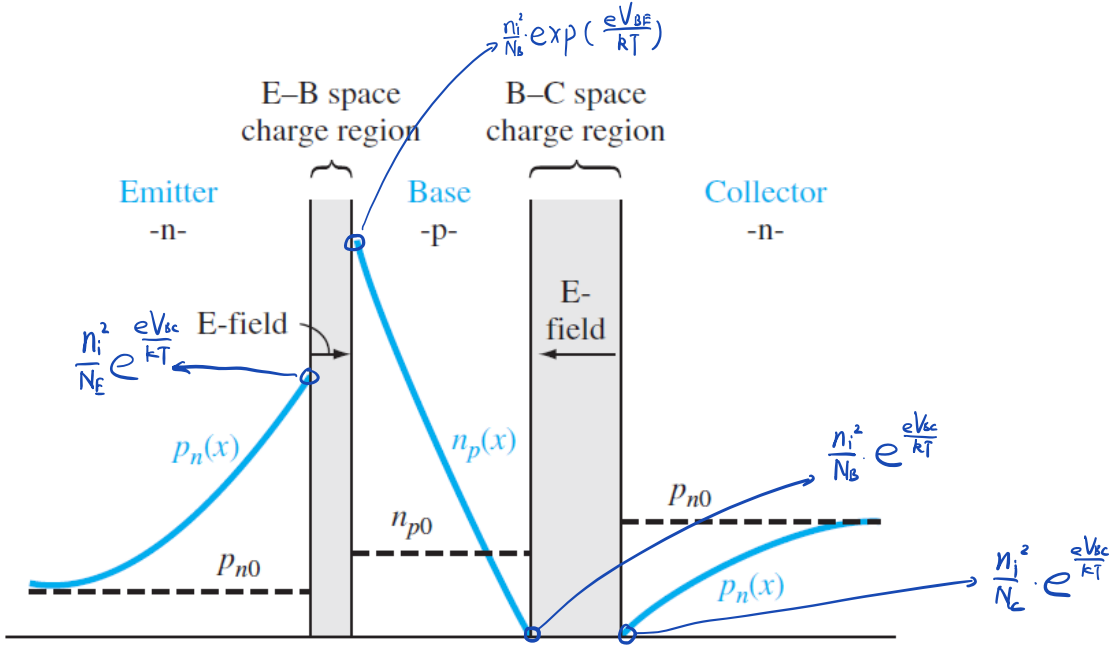
$\delta n_p = Ae^{\frac{x}{L_n}}+Be^{-\frac{x}{L_n}}$ $\quad L_n=\sqrt{D_n\tau_n}$

$\delta n_p(x)=\delta n_p(x_B)\cdot\frac{x}{x_B}+\delta n_p(0)\cdot\frac{x_B-x}{x_B}$

$J_n=eD_n\cdot\frac{\partial\delta n_p(x)}{\partial x}=-eD_n\cdot\frac{\delta n_p(0)-\delta n_p(x_B)}{x_B}$

Basic Principle of Operation



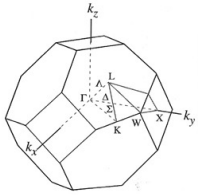


$J_E = J_{nE} + J_{pE}$ $J_{nE} \approx J_{pE} \approx J_C$. $J_B \approx J_{pE}$, no G&R

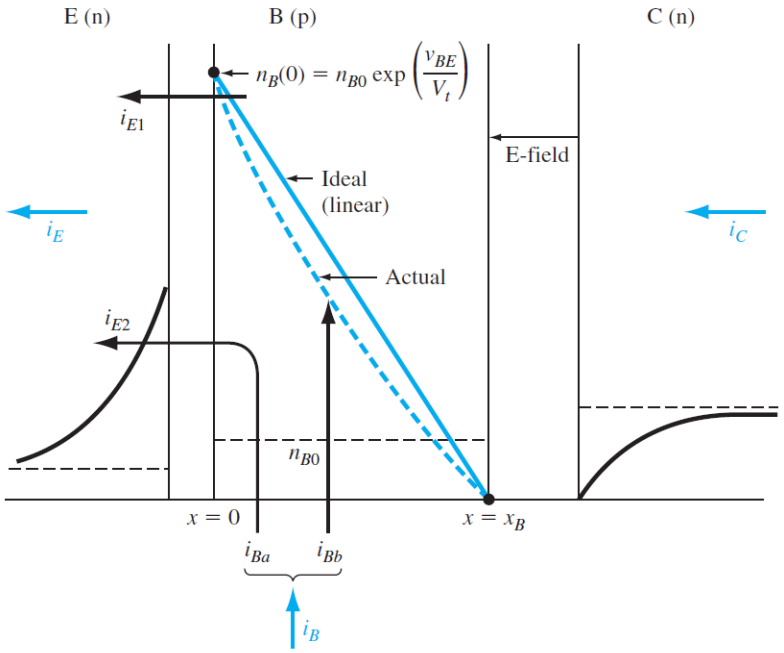
$\Rightarrow J_E = J_B + J_C$

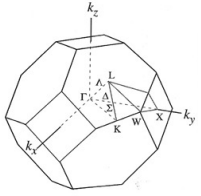
if $J_B \ll J_C$. $\beta = \frac{J_C}{J_B} \approx \frac{J_{nE}}{J_{pE}} \gg 1$

Common emitter current gain β.



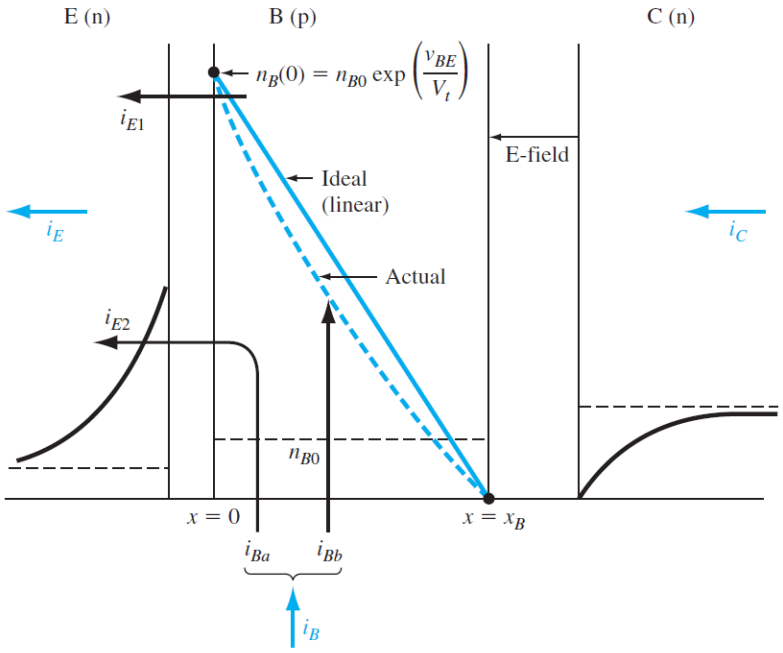
Collector Current





$$i_C = eD_nA_{BE}\frac{dn(x)}{dx} = eD_nA_{BE}\left[\frac{n_B(0) - 0}{0 - x_B}\right] = \frac{-eD_nA_{BE}}{x_B} \cdot n_{B0}\exp\left(\frac{v_{BE}}{V_t}\right)$$

Emitter Current

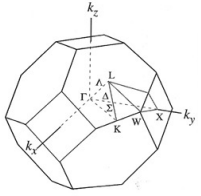


$$\frac{i_C}{i_E} \equiv \alpha$$

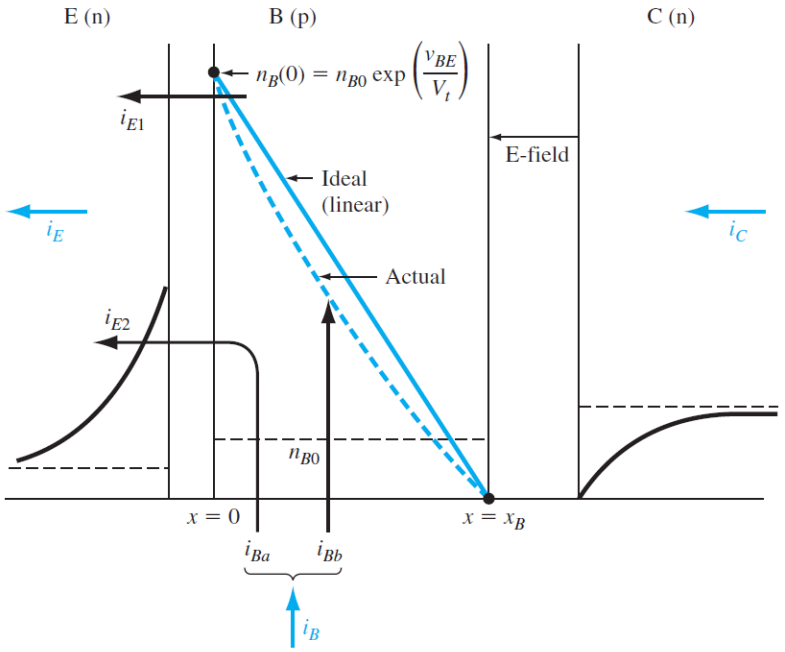
Common-base current gain

$$i_{E2} = I_{S2} \exp\left(\frac{v_{BE}}{V_t}\right)$$

$$i_E = i_{E1} + i_{E2} = i_C + i_{E2} = I_{SE} \exp\left(\frac{v_{BE}}{V_t}\right)$$



Base Current



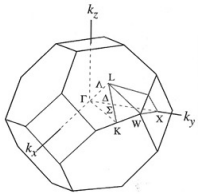
$$\frac{i_C}{i_B} \equiv \beta$$

Common-emittor current gain

$$\beta \approx \frac{J_{nE}}{J_{pE}}$$

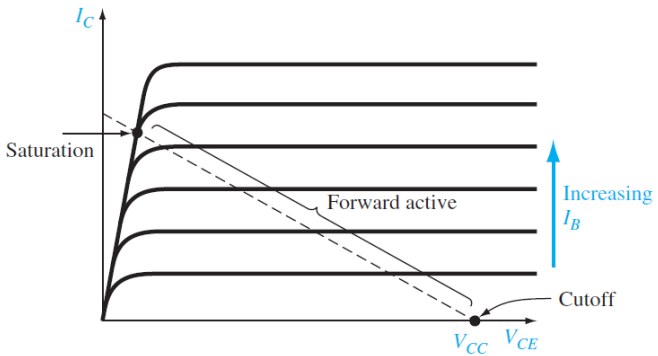
$$i_B = i_{Ba} + i_{Bb} = i_{E2} + i_{Bb}$$

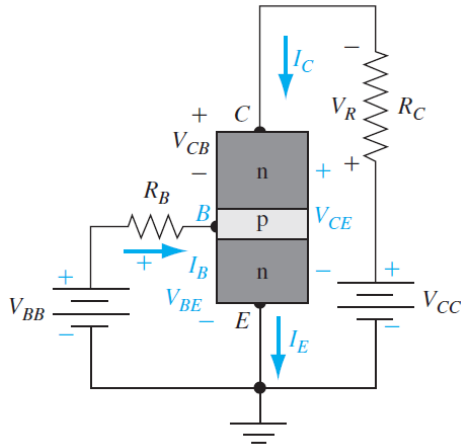
$$i_{Bb} \propto \delta n_p \propto e^{\frac{v_{BE}}{V_t}} - 1 \propto e^{\frac{v_{BE}}{V_t}}$$

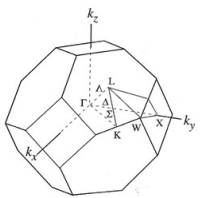


Modes of Operation

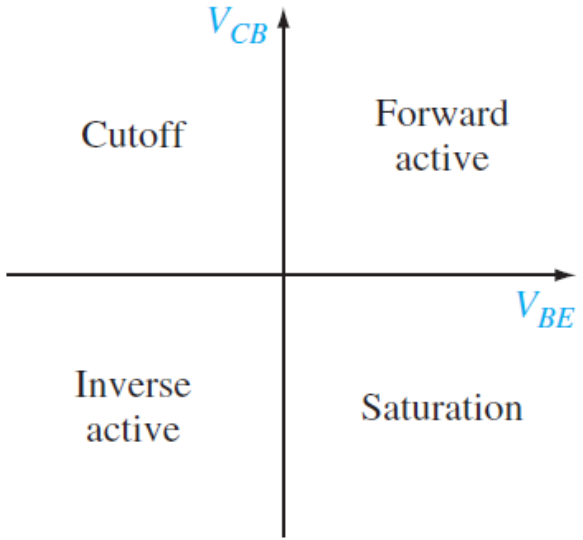
- Cutoff: $V_{BE} \leq 0, V_{BC} \leq 0$
- Forward active: $V_{BE} > 0, V_{BC} < 0$
- Saturation: $V_{BE} > 0, V_{CB} < 0$
- Inverse active: $V_{BE} \leq 0, V_{BC} > 0$

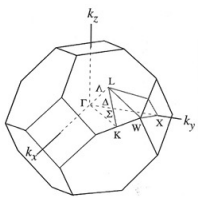






Modes of Operation





Amplification with Bipolar Transistors

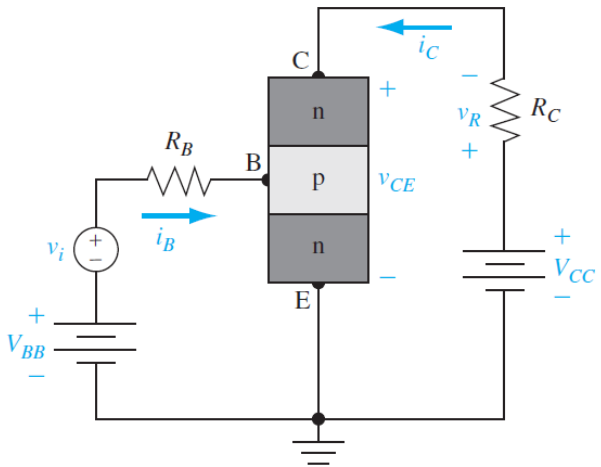
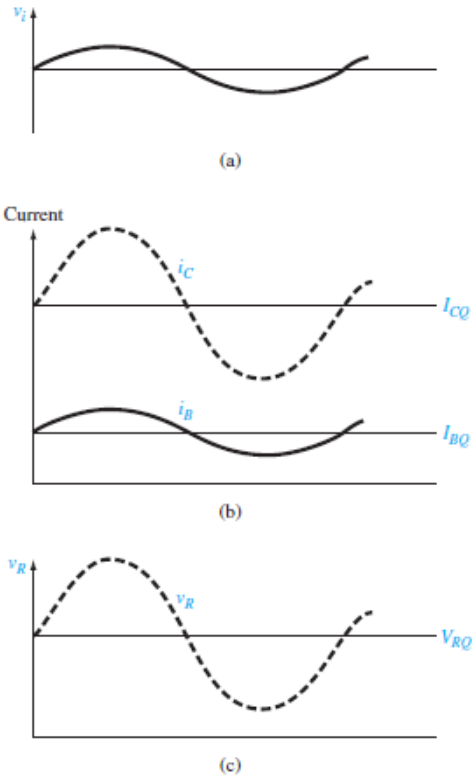
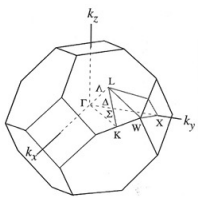


Figure 12.11 | Common-emitter npn bipolar circuit configuration with a time-varying signal voltage v_i included in the base–emitter loop.





Minority Carrier Distribution

- Forward-active mode

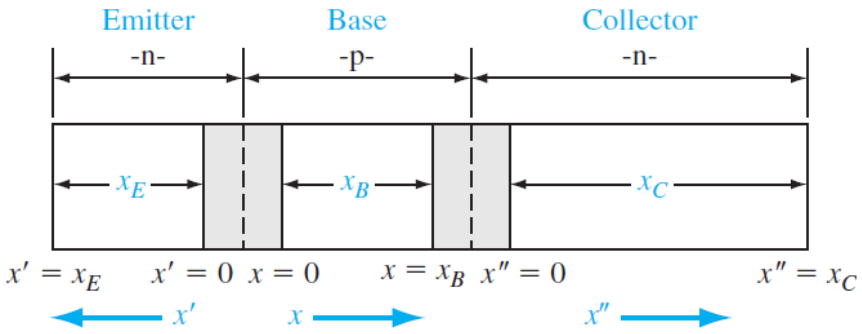


Figure 12.13 | Geometry of the npn bipolar transistor used to calculate the minority carrier distribution.

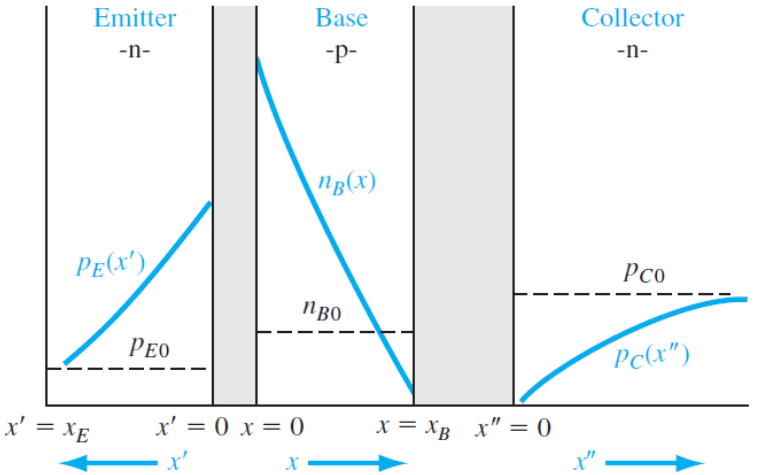
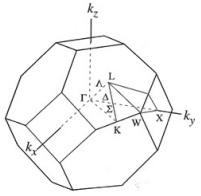


Figure 12.14 | Minority carrier distribution in an npn bipolar transistor operating in the forward-active mode.



Base Region

- Forward-active mode

$$D_B \frac{\partial^2(\delta n_B(x))}{\partial x^2} - \frac{\delta n_B(x)}{\tau_{B0}} = 0$$

$$\delta n_B(x) = A \exp\left(\frac{+x}{L_B}\right) + B \exp\left(\frac{-x}{L_B}\right)$$

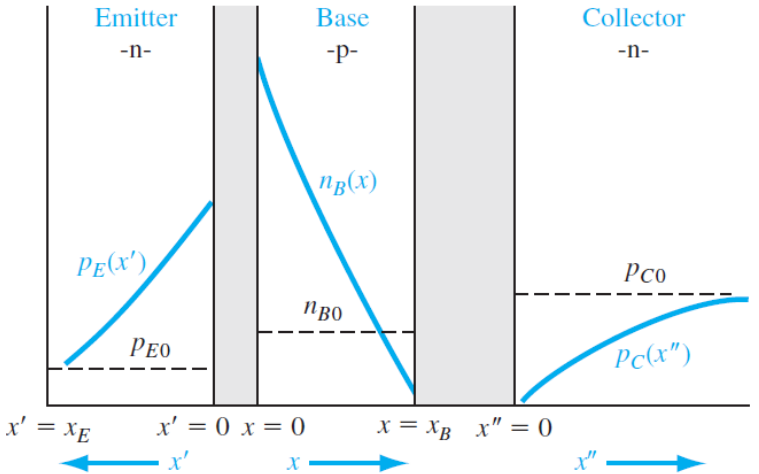
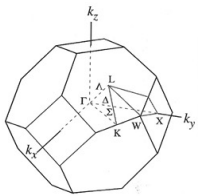


Figure 12.14 | Minority carrier distribution in an npn bipolar transistor operating in the forward-active mode.



Base Region

- Forward-active mode

$$\delta n_B(0) = n_B(x=0) - n_{B0} = n_{B0}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]$$

$$\delta n_B(x_B) = n_B(x=x_B) - n_{B0} = 0 - n_{B0} = -n_{B0}$$

$$\delta n_B(x) = \frac{n_{B0}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]\sinh\left(\frac{x_B - x}{L_B}\right) - \sinh\left(\frac{x}{L_B}\right)\right\}}{\sinh\left(\frac{x_B}{L_B}\right)}$$

$$\delta n_B(x) \approx \frac{n_{B0}}{x_B}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right](x_B - x) - x\right\}$$

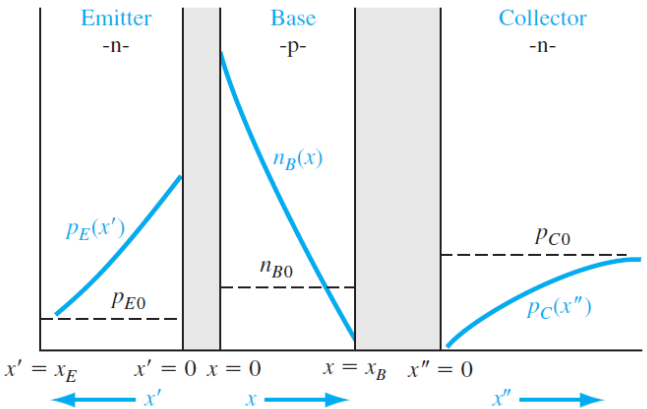
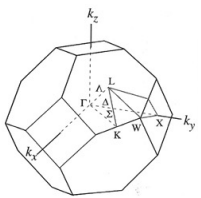


Figure 12.14 | Minority carrier distribution in an npn bipolar transistor operating in the forward-active mode.



Emitter Region

$$D_E \frac{\partial^2[\delta p_E(x')]}{\partial x'^2} - \frac{\delta p_E(x')}{\tau_{E0}} = 0$$

$$\delta p_E(x') = \frac{p_{E0}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]\sinh\left(\frac{x_E - x'}{L_E}\right)}{\sinh\left(\frac{x_E}{L_E}\right)}$$

$$\delta p_E(x') \approx \frac{p_{E0}}{x_E}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right](x_E - x')$$

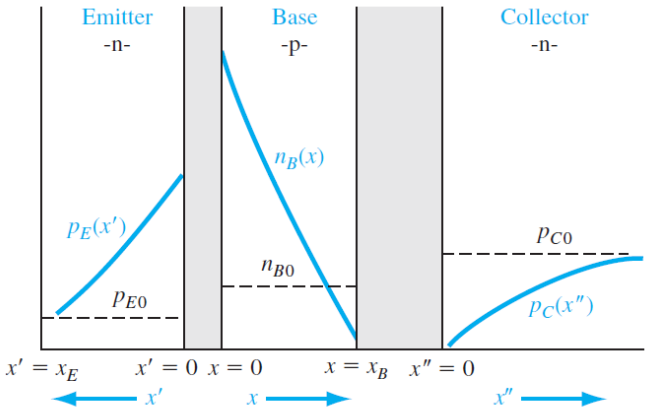
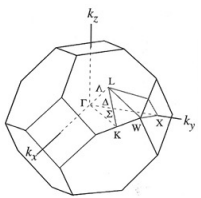


Figure 12.14 | Minority carrier distribution in an npn bipolar transistor operating in the forward-active mode.



Collector Region

$$D_C \frac{\partial^2[\delta p_C(x'')]}{\partial x''^2} - \frac{\delta p_C(x'')}{\tau_{C0}} = 0$$

$$\delta p_C(x'') = -p_{C0} \exp\left(\frac{-x''}{L_C}\right)$$

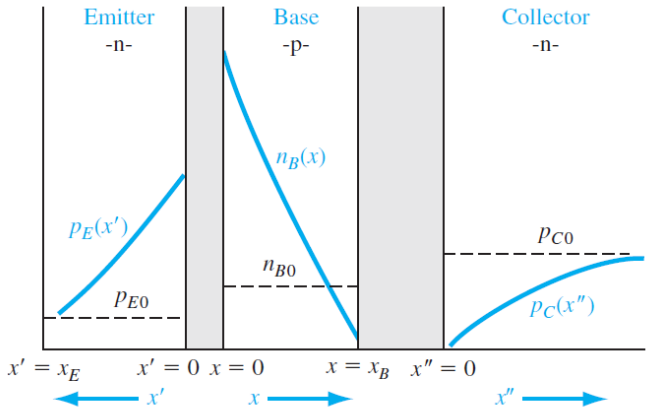
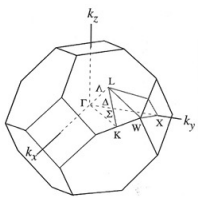
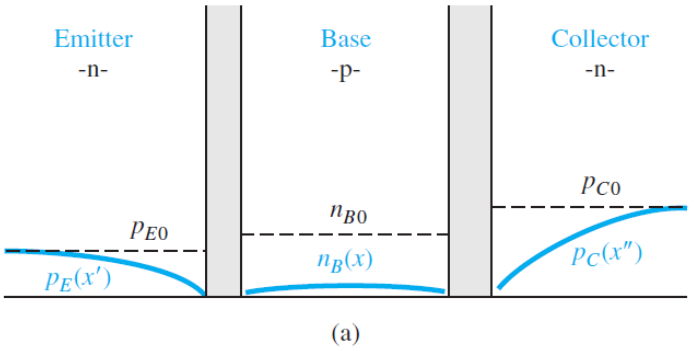


Figure 12.14 | Minority carrier distribution in an npn bipolar transistor operating in the forward-active mode.

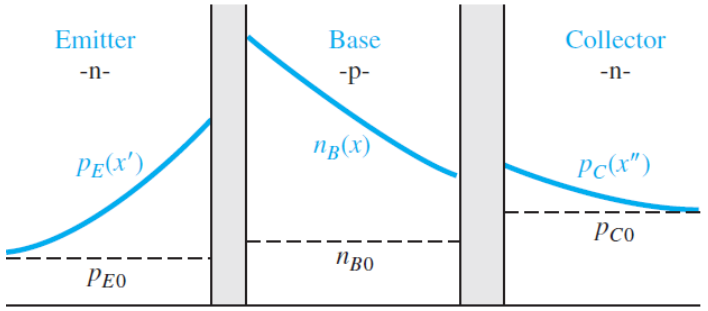


Other Operation Mode



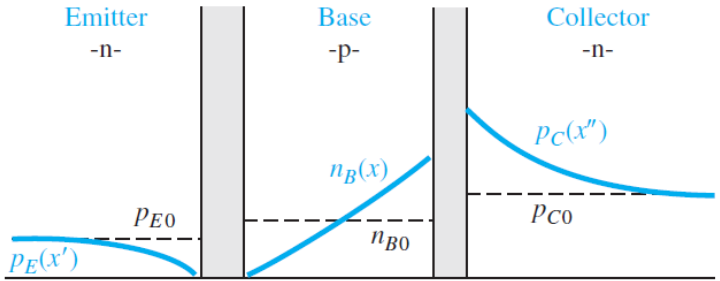
(a)

Cutoff

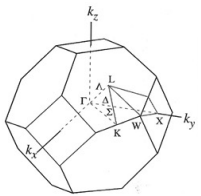


(b)

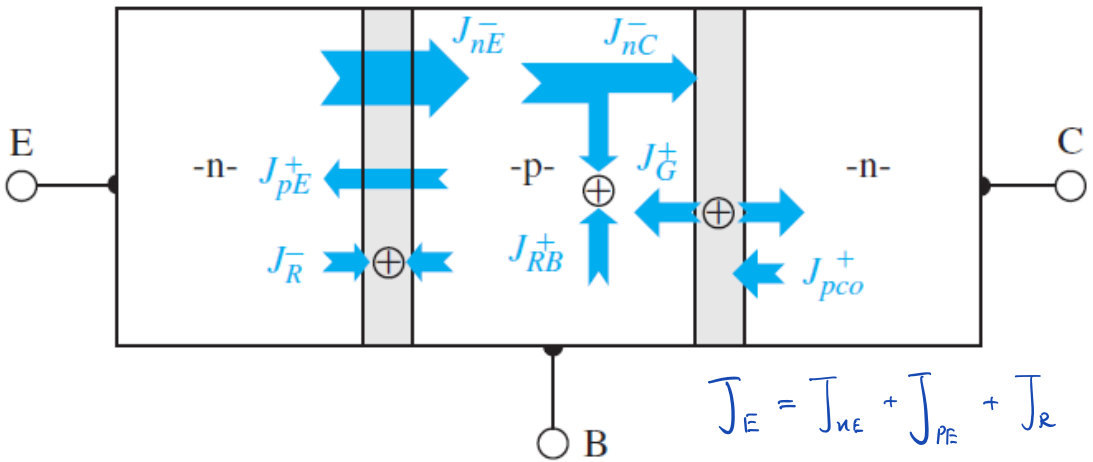
Saturation

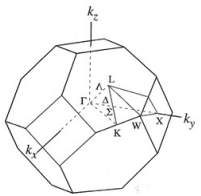


Inverse-active



Transistor Currents





Transistor Currents

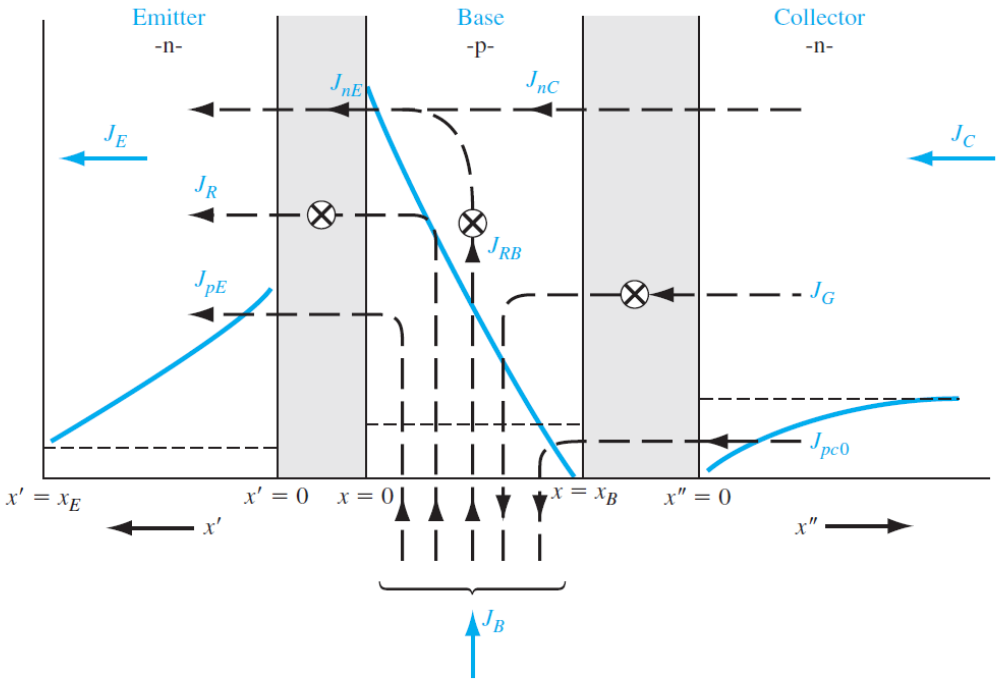
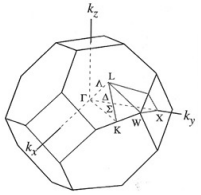


Figure 12.19 | Current density components in an npn bipolar transistor operating in the forward-active mode.



Common-base Current Gain

- DC common-base current gain:

$$\alpha_0 = \frac{J_C}{J_E} = \frac{J_{nC} + J_G + J_{pc0}}{J_{nE} + J_R + J_{pE}}$$

- Small signal current gain:

$$\alpha = \frac{\partial J_C}{\partial J_E} = \frac{J_{nC}}{J_{nE} + J_R + J_{pE}}$$

- $$\alpha = \left(\frac{J_{nE}}{J_{nE} + J_{pE}}\right)\left(\frac{J_{nC}}{J_{nE}}\right)\left(\frac{J_{nE} + J_{pE}}{J_{nE} + J_R + J_{pE}}\right)$$

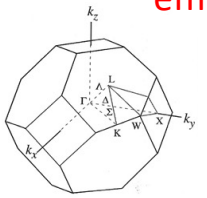
Emitter injection efficiency factor

Base transport factor

no G & R. = 1

with G & R. ≠ 1

Recombination factor



Common Emitter Current Gain

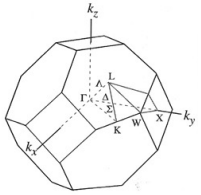
$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

$$\frac{1}{\alpha_0} = \frac{1}{\beta_0} + 1$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\alpha = \frac{\beta}{1 + \beta}$$

$$\alpha = \left(\frac{J_{nE}}{J_{nE} + J_{pE}}\right)\left(\frac{J_{nC}}{J_{nE}}\right)\left(\frac{J_{nE} + J_{pE}}{J_{nE} + J_R + J_{pE}}\right)$$



Emitter Injection Efficiency Factor

$$\gamma = \left(\frac{J_{nE}}{J_{nE} + J_{pE}}\right) = \frac{1}{\left(1 + \frac{J_{pE}}{J_{nE}}\right)}$$

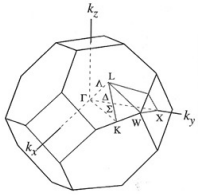
$$J_{pE} = -eD_E \left.\frac{d[\delta p_E(x')]}{dx'}\right|_{x'=0}$$

$$\delta p_E(x') \approx \frac{p_{E0}}{x_E}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right](x_E - x')$$

$$J_{nE} = (-)eD_B \left.\frac{d[\delta n_B(x)]}{dx}\right|_{x=0}$$

$$\delta n_B(x) \approx \frac{n_{B0}}{x_B}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right](x_B - x) - x\right\}$$

$$\gamma \approx \frac{1}{1 + \frac{N_B}{N_E} \cdot \frac{D_E}{D_B} \cdot \frac{x_B}{x_E}}$$



Base Transport Factor

$$J_{nC} = (-)eD_B \left.\frac{d[\delta n_B(x)]}{dx}\right|_{x=x_B}$$

$$\delta n_B(x) = \frac{n_{B0}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]\sinh\left(\frac{x_B - x}{L_B}\right) - \sinh\left(\frac{x}{L_B}\right)\right\}}{\sinh\left(\frac{x_B}{L_B}\right)}$$

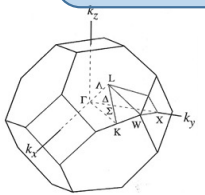
$$J_{nE} = (-)eD_B \left.\frac{d[\delta n_B(x)]}{dx}\right|_{x=0}$$

$$\delta n_B(x) \approx \frac{n_{B0}}{x_B}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right](x_B - x) - x\right\}$$

$$J_{nC} = \frac{eD_B n_{B0}}{L_B}\left\{\frac{[\exp(eV_{BE}/kT) - 1]}{\sinh(x_B/L_B)} + \frac{1}{\tanh(x_B/L_B)}\right\}$$

$$\alpha_T = \frac{J_{nC}}{J_{nE}} \approx \frac{\exp(eV_{BE}/kT) + \cosh(x_B/L_B)}{1 + \exp(eV_{BE}/kT)\cosh(x_B/L_B)}$$

$$\alpha_T \approx \frac{1}{\cosh(x_B/L_B)} \approx \frac{1}{1 + \frac{1}{2}(x_B/L_B)^2} \approx 1 - \frac{1}{2}(x_B/L_B)^2$$



Recombination Factor

$$\delta = \frac{J_{nE} + J_{pE}}{J_{nE} + J_R + J_{pE}} \approx \frac{J_{nE}}{J_{nE} + J_R} = \frac{1}{1 + J_R/J_{nE}}$$

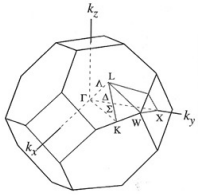
$$J_{pE} \ll J_{nE}$$

Why?

$$J_R = \frac{ex_{BE}n_i}{2\tau_0} \exp\left(\frac{eV_{BE}}{2kT}\right) = J_{r0} \exp\left(\frac{eV_{BE}}{2kT}\right)$$

$$J_{nE} = J_{s0} \exp\left(\frac{eV_{BE}}{kT}\right)$$

$$\delta = \frac{1}{1 + \frac{J_{r0}}{J_{s0}} \exp\left(\frac{-eV_{BE}}{2kT}\right)}$$



Summary of Limit Factors

Emitter injection efficiency

$$\gamma \approx \frac{1}{1 + \frac{N_B}{N_E} \cdot \frac{D_E}{D_B} \cdot \frac{x_B}{x_E}} \qquad (x_B \ll L_B),\ (x_E \ll L_E)$$

Base transport factor

$$\alpha_T \approx \frac{1}{1 + \frac{1}{2}\left(\frac{x_B}{L_B}\right)^2} \qquad (x_B \ll L_B)$$

Recombination factor

$$\delta = \frac{1}{1 + \frac{J_{r0}}{J_{s0}} \exp\left(\frac{-eV_{BE}}{2kT}\right)}$$

Common-base current gain

$$\alpha = \gamma \alpha_T \delta \approx \frac{1}{1 + \frac{N_B}{N_E} \cdot \frac{D_E}{D_B} \cdot \frac{x_B}{x_E} + \frac{1}{2}\left(\frac{x_B}{L_B}\right)^2 + \frac{J_{r0}}{J_{s0}} \exp\left(\frac{-eV_{BE}}{2kT}\right)}$$

Common-emitter current gain

$$\beta = \frac{\alpha}{1 - \alpha} \approx \frac{1}{\frac{N_B}{N_E} \cdot \frac{D_E}{D_B} \cdot \frac{x_B}{x_E} + \frac{1}{2}\left(\frac{x_B}{L_B}\right)^2 + \frac{J_{r0}}{J_{s0}} \exp\left(\frac{-eV_{BE}}{2kT}\right)}$$

